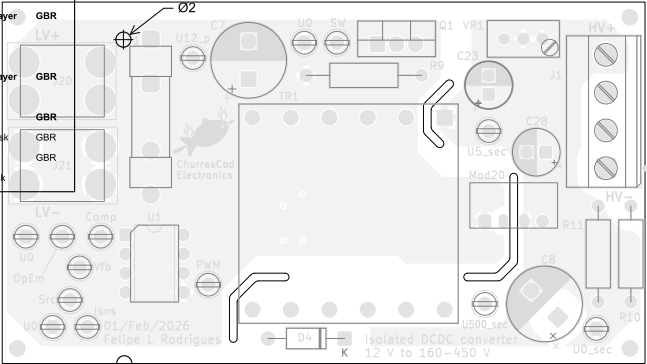


Module Fabrication Document

Layer Stack Legend

	Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste				Paste Mask	
	F.Silkscreen			Direct Printing	Legend	GBR
	F.Mask		0.02mm	Solder Resist	Solder Mask	GBR
	Copper	L1 (Ref)	0.07mm (2.00oz)		Plane	GBR
	Prepreg		0.1mm	FR4_7628	Dielectric	
	Copper	Virtual (Top side jumpers)	0.035mm (1.00oz)		Copper Layer	GBR
	Core		1.21mm	FR4	Dielectric	
	Copper	Virtual (Bot side jumpers)	0.035mm (1.00oz)		Copper Layer	GBR
	Prepreg		0.1mm	FR4	Dielectric	
	Copper	L2 (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
	B.Mask		0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen			Direct Printing	Legend	GBR
	B.Paste				Paste Mask	
Total thickness: 1.66mm						
Note: external layer thicknesses are specified after plating						

Top Fabrication (Scale 1:1)



Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
NA	-	-	-	-	-	-

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- FABRICATE PER IPC-6012A CLASS 2.
- OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- SURFACE FINISH: IMMERSION GOLD
- SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR TOP: #8071501A / BOTTOM: GREEN.
- SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- PCB MATERIAL REQUIREMENTS:
 - FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
 - Tg 170 C OR EQUIVALENT.
 - EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY FR.
- DESIGN GEOMETRY MINIMUM FEATURE SIZES:

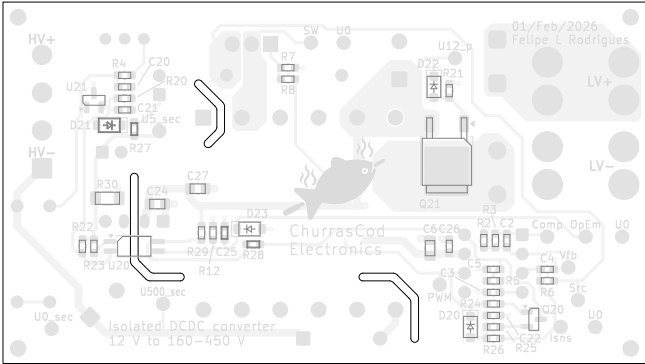
BOARD SIZE	85.000 × 47.800 mm
BOARD THICKNESS	1.660 mm
TRACE WIDTH	0.400 mm
TRACE TO TRACE	-0.000 mm
MIN. HOLE (PTH)	0.250 mm
MIN. HOLE (NPTH)	2.100 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm

All dimensions are in millimeters unless otherwise specified.

	Comments: Comment 1 Comment 2 Comment 3 Comment 4	Company: FR	Variant: PRELIMINARY	Git Hash: da3196a
		Board Name: Module	Project Name: Isolated step-up 12:450 V	
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2024-04-13 Revision: 1.0.0+ (Unreleased)
	Sheet Path:	Reviewer: NA	Size: A4	Sheet: 1 of 10

Module Fabrication Document

Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

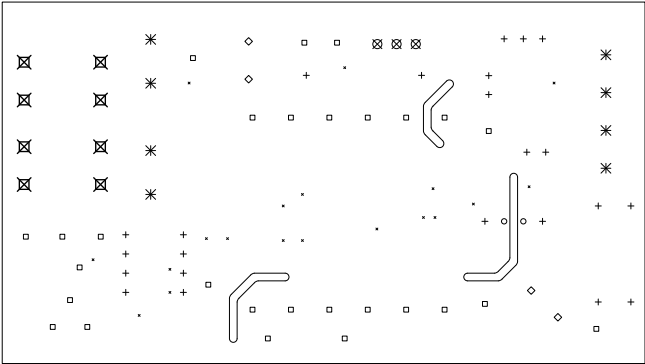
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	Comment 2	Board Name: Module		Project Name:	
	Comment 3			Isolated step-up 12:450 V	
	Comment 4				
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Bottom Fabrication (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
	Sheet Path:		Reviewer:	Size:	Sheet:
			NA	A4	2 of 10

Module Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	19	0,25mm (9,84mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
○	2	0,70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
+	23	0,80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
□	28	0,90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
◇	4	1,00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
⊠	3	1,20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
✱	8	1,30mm (51,18mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
⊞	8	1,80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
Total 95						

Drill Drawing L1 - L4 (Scale 1:1)



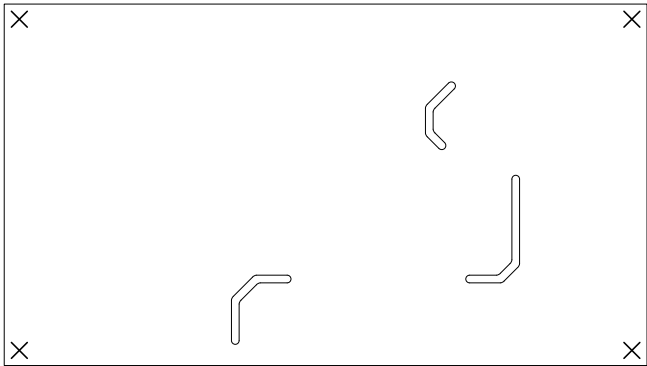
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	Comment 4	Module		Isolated step-up 12:450 V	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Drill Drawing (L1 - L4)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
	Sheet Path:		Reviewer:	Size:	Sheet:
			NA	A4	3 of 10

Module Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	2.10mm (82.68mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
	Total 4					

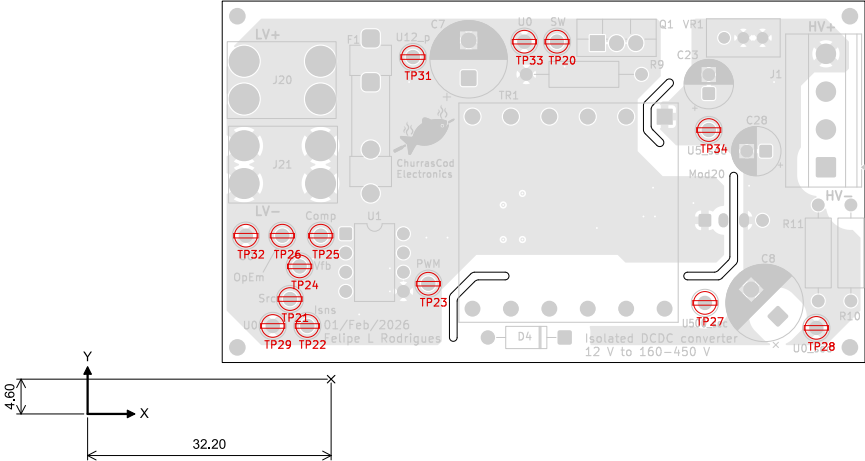
Drill Drawing L1 - L4 (Scale 1:1)



	Comments:	Company:		Variant:	Git Hash:
	Comment 1	FR		PRELIMINARY	da3196a
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	Comment 3	Module		Isolated step-up 12:450 V	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Drill Drawing (L1 - L4)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
	Sheet Path:		Reviewer:	Size:	Sheet:
			NA	A4	4 of 10

Module Fabrication Document

Top Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP20	SW	62.07	49.23
TP21	Src	26.77	15.19
TP22	Isns	29.05	11.64
TP23	PWM	45.05	17.22
TP24	Vfb	28.04	19.51
TP25	Comp	30.83	23.57
TP26	OpEm	25.75	23.57
TP27	U500_sec	81.63	14.68
TP28	U0_sec	96.36	11.38
TP29	U0	24.48	11.64
TP31	U12_p	43.02	47.20
TP32	U0	20.92	23.57
TP33	U0	57.75	49.23
TP34	U5_sec	82.14	37.54

Ref.	Net	X [mm]	Y [mm]
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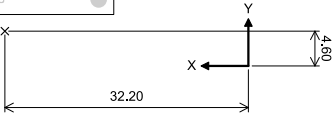
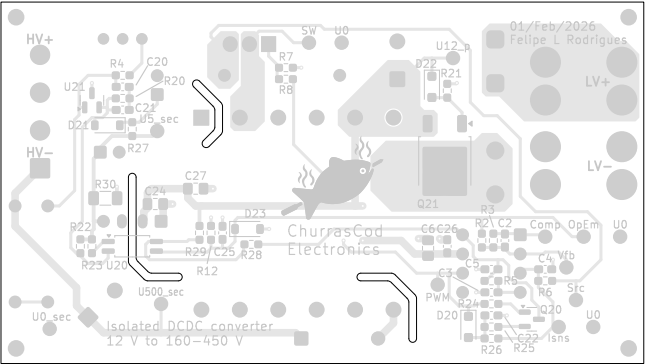
All dimensions are in millimeters unless otherwise specified.

	Comments:	Company:		Variant:	Git Hash:
	Comment 1	FR		PRELIMINARY	da3196a
	Comment 2	Board Name:		Project Name:	
	Comment 3				
	Comment 4	Module		Isolated step-up 12:450 V	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Top Test Points (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
	Sheet Path:		Reviewer:	Size:	Sheet:
			NA	A4	5 of 10

Module Fabrication Document

Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
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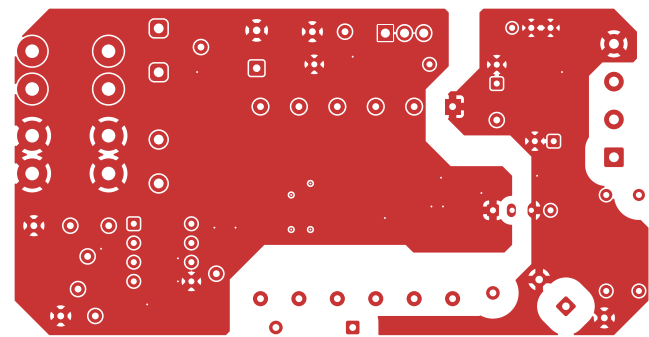


All dimensions are in millimeters unless otherwise specified.

	Comments: Comment 1 Comment 2 Comment 3 Comment 4	Company: FR		Variant: PRELIMINARY	Git Hash: da3196a
		Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Bottom Test Points (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
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Module Fabrication Document

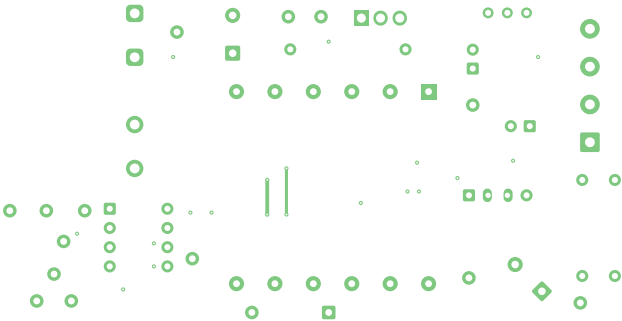
L1 (Ref) (Scale 1:1)



	Comments:	Company:		Variant:	Git Hash:
	Comment 1	FR		PRELIMINARY	da3196a
	Comment 2	Board Name:		Project Name:	
	Comment 3	Module		Isolated step-up 12:450 V	
	Comment 4	File Name:	Designer:	Date:	Revision:
	L1 (Ref) (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
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		NA	A4	7 of 10	

Module Fabrication Document

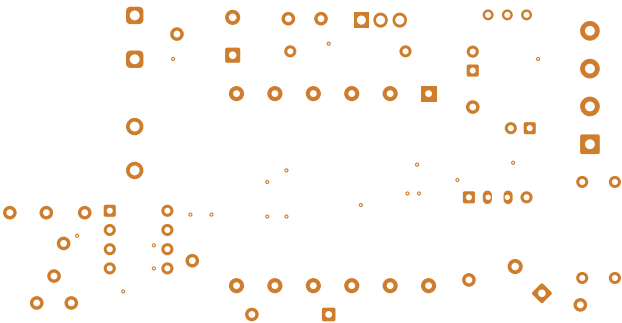
Virtual (Top side jumpers) (Scale 1:1)



	Comments:	Company:		Variant:	Git Hash:
	Comment 1	FR		PRELIMINARY	da3196a
	Comment 2	Board Name:		Project Name:	
	Comment 3	Module		Isolated step-up 12:450 V	
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		Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
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	Virtual (Top side jumpers) (Scale 1:1)	NA		A4	8 of 10
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Module Fabrication Document

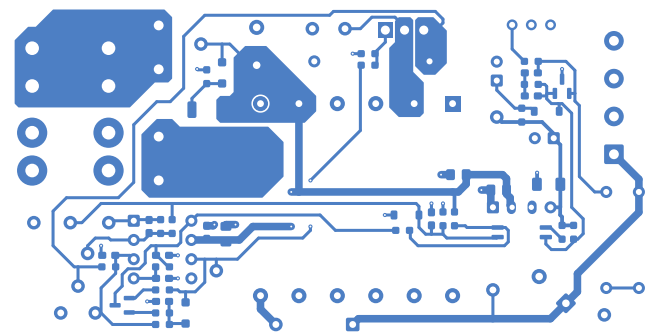
Virtual (Bot side jumpers) (Scale 1:1)



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	Comment 3	Module		Isolated step-up 12:450 V	
	Comment 4	File Name:		Date:	Revision:
		Isolated_12to450V.kicad_pcb		2024-04-13	1.0.0+ (Unreleased)
	Sheet Title:	Designer:		Size:	Sheet:
	Virtual (Bot side jumpers) (Scale 1:1)	FR		A4	9 of 10
	Sheet Path:	Reviewer:			
		NA			

Module Fabrication Document

L2 (Sig, PWR) (Scale 1:1)



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	Comment 1	FR		PRELIMINARY	da3196a
	Comment 2	Board Name:		Project Name:	
	Comment 3	Module		Isolated step-up 12:450 V	
	Comment 4	File Name:		Date:	Revision:
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	Sheet Title:	Designer:		Size:	Sheet:
	L2 (Sig, PWR) (Scale 1:1)	FR		A4	10 of 10
	Sheet Path:	Reviewer:			
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